

Bias and Hyperparameter tuning

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Bias % It refers to the systematic error that can affect the patterns discovered by an unsupervised model.

Eg: predict House price, but model only looks at house size and ignores location, number of rooms etc

High bias $\rightarrow$ underfitting
Model is too simple

Low Bias $\rightarrow$ Better learning
Model can capture more patterns from the data.

Hyperparameter tuning % It means finding the best settings for a clustering algorithm before training.

Examples %

KNN $\rightarrow$ K is Hyperparameter tuning
Decision tree $\rightarrow$ max-depth

Random Forest $\rightarrow$ n_estimators

SVM $\rightarrow$ C, kernel

k-Means $\rightarrow$ k

Goal % Find the best hyperparameter for better generalization.